

Data Science Essentials

08



Data science turns raw data into powerful insights, revealing patterns and trends that drive smarter decisions and innovations in today's data-driven world.



Data is everywhere around us. From the moment we wake until we go to bed, we interact with data in countless ways. For instance, when you browse the internet, every click, search, and website visit is recorded as data. Even your daily routine, such as the time you leave for your school or the route you take, can be considered data.

Data is a collection of information gathered through observation, measurement, or questions. It can be numbers, words, website clicks, or even the temperature outside.

Data Analysis

The data, when collected and inspected, can reveal patterns, trends, and insights that help us to make informed decisions. For example, a business may analyse sales data to determine which products are most popular at certain times of the year. This method of studying or analysing data to find useful patterns is called **data analysis**.

Industries and businesses use data analysis to understand customer preferences, improve products, and make better decisions. However, data analysis has its limitations.

Consider an example of a store that tracks which products are most frequently purchased. Data analysis may help the store identify trends and determine which items are popular. However, it cannot help the store predict new products that customers might prefer, recommend items based on individual preferences, or forecast future sales. To perform such advanced tasks, data science is required.

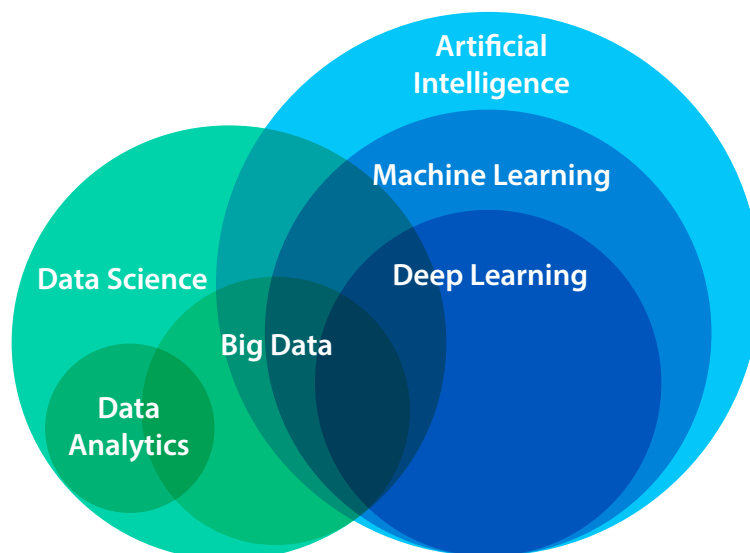


Data Science

Data science is the study of data to extract meaningful insights and make predictions that support informed decision-making. It involves collecting, analysing, and interpreting large amounts of data using statistics, domain knowledge and programming languages.



By combining data analysis with advanced methods of artificial intelligence, data science enables more accurate predictions and personalised recommendations. Data science is a broad field that encompasses aspects of mathematics, statistics, data analytics, and Artificial Intelligence.



Applications of Data Science

Data science plays a vital role in helping businesses solve problems and make informed decisions. It is applied in various fields, such as healthcare, finance, marketing, and transportation, to enhance services and optimise operations. Let us look at a few applications.

1. **Healthcare:** Data science predicts disease outbreaks and tailor treatments by analysing patient and genetic data.
2. **Marketing:** It helps target ads to specific audiences and evaluates the effectiveness of marketing campaigns.
3. **Transportation:** Data science studies traffic patterns and predicts when public transportation systems need maintenance.
4. **Finance:** It detects fraud by identifying unusual transactions and forecasts stock market trends to guide investment decisions.

Data Science Life Cycle

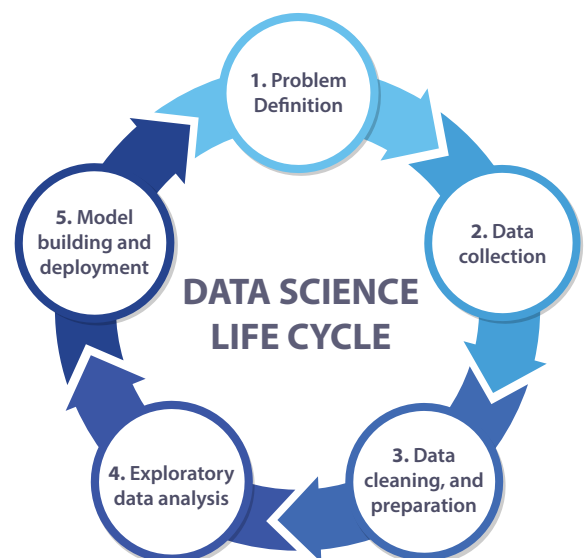
In data science, managing and analysing large volumes of complex data can lead to inefficiencies and errors. A structured approach is essential for effectively managing a data science project. This is known as the **Data Science Life Cycle**, which guides you step-by-step from understanding the problem to building and deploying a model. Each step is crucial to ensure that the project yields valuable insights and actionable results.

Phases in Data Science Life Cycle

The core phases outlined in the figure are essential for structuring any data science project, from simple to moderately complex ones.

For more complex projects, additional steps such as feature engineering, model tuning, and validation are required.

Understanding each phase is crucial for project implementation. Let us explore each of these phases in detail.



Understanding Data Science Life Cycle

Scenario

A school recently opened a new sports club but noticed a surprisingly low number of enrollments. To address this, the school wants to analyse the reasons behind the low interest and identify students who may be inclined to join the club.

1. Define and Understand the Problem

The first step in the Data Science Life Cycle is to clearly **define the problem** you want to solve. This is crucial since it determines the whole project. Without a clear problem statement, you might gather irrelevant data or use incorrect methods, which can lead to wrong conclusions. Clearly outlining the problem helps set specific goals and objectives, making it easier to find the right solution.

In the school scenario, the initial objective is to identify the causes of the poor enrolment in the new sports club.



Problem Statement: Why is there such a low enrolment rate for the new sports club?

By clearly understanding and defining this issue, you can focus on determining the reasons behind the low enrolment rate and explore ways to encourage more students to participate.

2. Data Collection

In the Data Collection phase, you gather all the necessary information to address the problem. This data may originate from various sources, such as databases, surveys, or records. Collecting accurate and relevant data is crucial, as the quality and quantity of the data directly affect the analysis. If the data is incorrect or insufficient, you may be unable to solve the problem effectively.



Considering the school scenario, it is essential to collect relevant student data to determine the reasons behind the low enrolment in the sports club. This may include information on students' interests, engagement in other activities, and survey feedback. Analysing this data will enable the identification of patterns or issues that may be influencing their decision.

3. Data Cleaning and Preparation

Following data collection, the next step is to clean and prepare the data for analysis. This phase is critical since the data quality directly influences the accuracy of the analysis. Data cleaning involves rectifying errors, addressing missing values, and ensuring data consistency. Inaccuracies or incomplete data, if not properly addressed, can significantly affect the validity of the results.

Examining the school scenario, after collecting survey data from students, you may notice that some responses are incomplete or unclear. For example, some students might not have answered all the questions. These issues can be resolved by filling in the missing information or correcting any errors, making sure the dataset is ready for analysis.

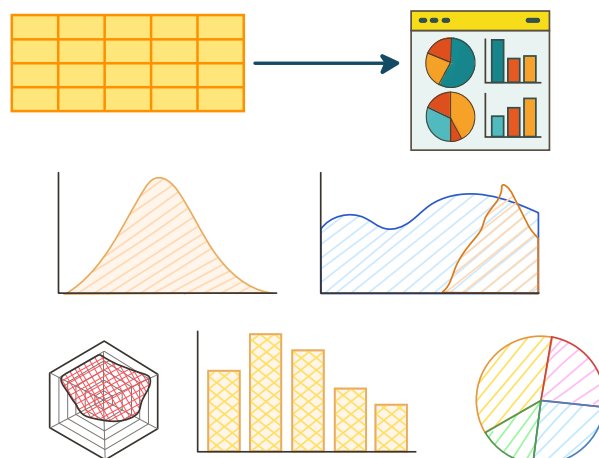
There are a few ways to handle missing data.

1. Filling in Missing Values:

- a. **Imputation:** Estimate the missing values based on other available data, like using the most common or average value.
 - b. **Default Values:** Assign a basic value like 'Unknown' or 'No' if the missing data cannot be accurately guessed.
2. **Removing Incomplete Data:** If only a small portion of data is missing and imputing might introduce inaccuracies, you may choose to delete those rows. However, be cautious as this approach reduces the overall dataset size.
 3. **Flagging Missing Data:** Add a new column to indicate where data was missing and subsequently addressed. This allows you to assess how these modifications might impact your analysis.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) involves examining data to identify patterns, trends, and insights. During this phase, visualisations and basic statistical methods are used to gain a deeper understanding of the data. EDA gives an early look at the data, helping to spot issues or interesting patterns before building models.



With respect to the school scenario, after cleaning the data, you may observe a trend where students already engaged in other activities are less likely to join the sports club. Additionally, you might find that students with a strong interest in sports are more inclined to sign up. These insights, gained through Exploratory Data Analysis (EDA), can help identify the key factors affecting signup rates and guide strategies to address them.

5. Model Building and Deployment

In this final phase, the insights gained from EDA are used to build and train a model that can make predictions based on the data. After training, you evaluate the model to determine its accuracy before deploying it for real-world use. This phase turns the insights gathered into actionable decisions.

In the school scenario, based on the analyzed data, you can create a simple machine learning model to identify which students are likely to join the sports club in the future. The model could consider factors such as their interest in sports, involvement in other activities, and time availability. By deploying this model, you can better target your efforts to encourage students who are more likely to join, ultimately increasing the club's signup rate.

Now, we will perform the technical activity to understand these phases better.

Activity

Let us implement data science in a scenario where a company aims to determine which marketing channel—newspaper, TV, or radio—is most effective in boosting sales.

Part 1 - Define the Problem

The first step is to understand the problem and define it. This helps in setting the goal and course of the project.

a. Understand the Goal:

Our goal is to analyse how marketing expenditures on TV, Radio, and Newspapers impact sales. We want to identify which type of advertising results in the highest increase in sales.

b. Specify the Objective:

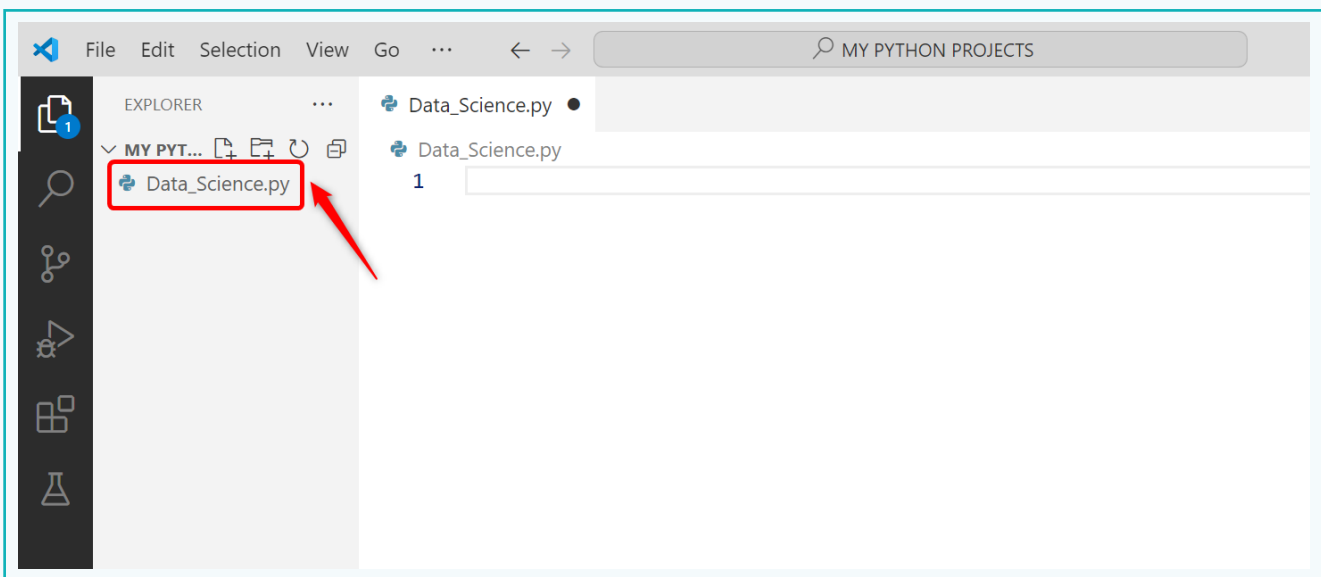
The objective is to create a model that predicts sales based on the amount spent on each marketing channel. This can help you make more informed decisions about where to allocate the marketing budget.

Activity

Part 2 - Collect the Data

1 Initial Setup

- Create a new folder in the desired location.
- Open the VS Code editor, click on the 'File' menu, select 'Open Folder'. Locate the folder you have created and click on 'Select Folder'.
- Once the folder is displayed in the VS Code editor sidebar, create a new file within it. Name the file "Data_Science.py".



Install necessary modules

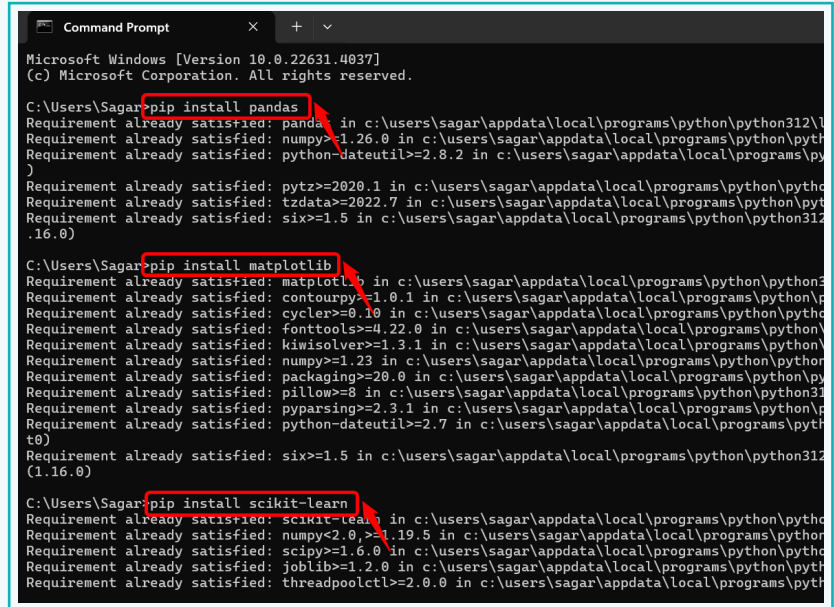
In data science, installing Python modules is essential to simplify data manipulation, visualisation, and modelling tasks.

- **Pandas:** A library that provides data structures like DataFrames, making it easier to handle and analyse data in Python.
- **Matplotlib:** A library for creating static, interactive, and animated visualizations in Python.
- **Linear Regression:** A Machine Learning algorithm used to identify the relationship between features and a target variable, helping in prediction and analysis.

Activity

- 2 Open the Command prompt and pass the following commands to install all three modules.

- `pip install pandas`
- `pip install matplotlib`
- `pip install scikit-learn`



```
Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

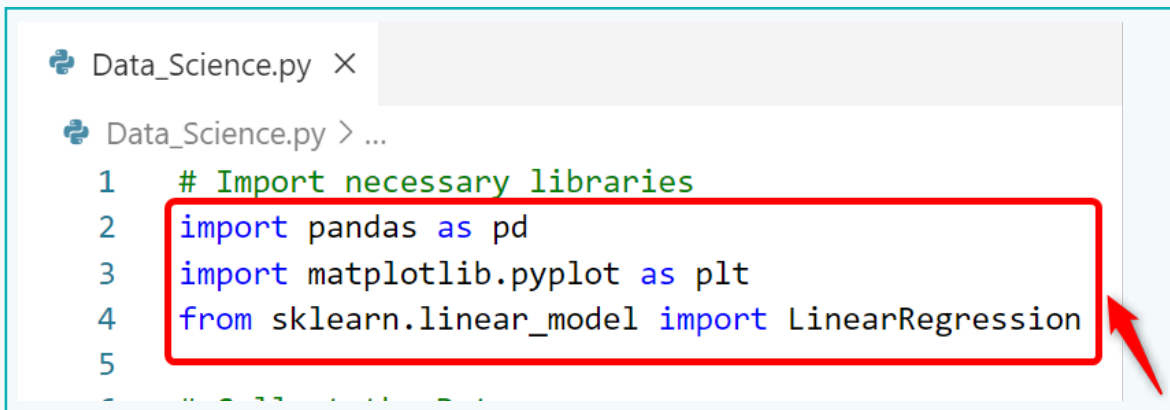
C:\Users\Sagar>pip install pandas
Requirement already satisfied: pandas in c:\users\sagar\appdata\local\programs\python\python312\
Requirement already satisfied: numpy>=1.26.0 in c:\users\sagar\appdata\local\programs\python\pyth
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\sagar\appdata\local\programs\py
)
Requirement already satisfied: pytz>=2020.1 in c:\users\sagar\appdata\local\programs\python\pyth
Requirement already satisfied: tzdata>=2022.7 in c:\users\sagar\appdata\local\programs\python\pyt
Requirement already satisfied: six>=1.5 in c:\users\sagar\appdata\local\programs\python\python312
(1.16.0)

C:\Users\Sagar>pip install matplotlib
Requirement already satisfied: matplotlib in c:\users\sagar\appdata\local\programs\python\python3
Requirement already satisfied: contourpy>=1.0.1 in c:\users\sagar\appdata\local\programs\python\p
Requirement already satisfied: cycler>=0.10 in c:\users\sagar\appdata\local\programs\python\pytho
Requirement already satisfied: fonttools>=4.22.0 in c:\users\sagar\appdata\local\programs\python\pytho
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\sagar\appdata\local\programs\python\pytho
Requirement already satisfied: numpy>=1.23 in c:\users\sagar\appdata\local\programs\python\pytho
Requirement already satisfied: packaging>=20.0 in c:\users\sagar\appdata\local\programs\python\py
Requirement already satisfied: pillow>=8 in c:\users\sagar\appdata\local\programs\python\python31
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\sagar\appdata\local\programs\python\p
Requirement already satisfied: python-dateutil>=2.7 in c:\users\sagar\appdata\local\programs\pyth
(t0)
Requirement already satisfied: six>=1.5 in c:\users\sagar\appdata\local\programs\python\python312
(1.16.0)

C:\Users\Sagar>pip install scikit-learn
Requirement already satisfied: scikit-learn in c:\users\sagar\appdata\local\programs\python\pytho
Requirement already satisfied: numpy<2.0, >=1.19.5 in c:\users\sagar\appdata\local\programs\pytho
Requirement already satisfied: scipy>=1.6.0 in c:\users\sagar\appdata\local\programs\python\pytho
Requirement already satisfied: joblib>=1.2.0 in c:\users\sagar\appdata\local\programs\python\pyth
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\sagar\appdata\local\programs\pyth
```

Import Libraries

- 3 Import the libraries required for data handling, visualisation, and modelling.
- Import *pandas* with the alias *pd* for efficient data manipulation.
 - Import *pyplot* from *matplotlib* with the alias *plt* for creating plots.
 - Import *LinearRegression* from *scikit-learn* for building and training the regression models.



```
Data_Science.py X
Data_Science.py > ...
1 # Import necessary libraries
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from sklearn.linear_model import LinearRegression
5
```

Now, we need to load the data from the Web.

Activity

- 4 Using Pandas, import the dataset from the given URL:
(<https://www.statlearning.com/s/Advertising.csv>).

This dataset includes information about advertising expenditures and corresponding sales.

- Save the website link as a string in the *url* variable.
- Use the *pd.read_csv()* function with parameters *url* and *index_col*. Specify *index_col=0* to set the first column as the index and save it in *df* variable.

```
6 # Collect the Data
7 url = 'https://www.statlearning.com/s/Advertising.csv'
8 df = pd.read_csv(url, index_col=0)
9
```

- 5 Display the first few rows of the data.

- Use the *head()* function to display the first few rows of the DataFrame to verify that the data has been loaded correctly.
- Wrap it in the *print()* function to display the output.

```
9
10 # Preview the Data
11 print(df.head())
12
```

Output -

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

>>> url = 'https://www.statlearning.com/s/Advertising.csv'
>>> df = pd.read_csv(url, index_col=0)
>>> print(df.head())
   TV  radio  newspaper  sales
1  230.1   37.8     69.2   22.1
2   44.5   39.3     45.1   10.4
3   17.2   45.9     69.3    9.3
4  151.5   41.3     58.5   18.5
5  180.8   10.8     58.4   12.9
>>>
```

Activity

Part 3 - Prepare the Data

1 Check for Missing Values

- Before processing the data, we must check for the missing values. For this, we will use the `isnull()` function.

`isnull()`: The `isnull()` function checks for missing values in the DataFrame.

Syntax - `df.isnull()`

`sum()`: The `sum()` function counts the number of missing values in each column.

Syntax - `df.sum()`

- Use the `isnull()` function combined with `sum()` to check for any missing values in the dataset. This will show how many missing values exist in each column.
- Wrap the functions in the `print()` function to display the output.

```
13 # Prepare the Data
14 # Check for missing values
15 print(df.isnull().sum())
16
```



Output -

```
>>> print(df.isnull().sum())
TV          0
radio       0
newspaper   0
sales       0
dtype: int64
```



Thus, we can see there are no null values present in the data. Therefore, we can proceed to EDA.

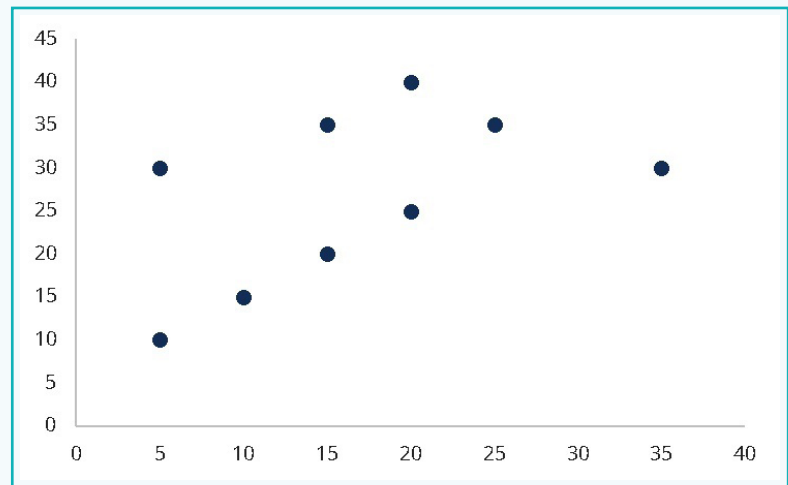
Activity

Part 4 - Exploratory Data Analysis (EDA)

Now let us visualise the relationships by plotting the information on a graph to gain a better understanding of the data. For this, we will use a scatter plot, as it helps in visualising the relationship between two continuous variables easily.

Scatter Plot

- A Scatter Plot is a type of data visualisation that uses dots to represent the values of two different variables.
- Each dot represents an observation from the dataset, with its position determined by the values of the two variables being compared.
- Scatter plots are useful for identifying relationships, trends, and correlations between variables.



Scatter plot

Syntax:

`plt.scatter(x, y)`: Creates a scatter plot with x and y as the data for the x-axis and y-axis, respectively.

`plt.xlabel()`: Sets the label for the x-axis.

`plt.ylabel()`: Sets the label for the y-axis.

`plt.title()`: Sets the title of the plot.

`plt.show()`: Displays the plot.

`Plt.figure()`: Sets a new figure for plotting, allowing customisation of the figure size and layout.

Now, create scatter plots to explore the relationship between marketing expenditures and sales.

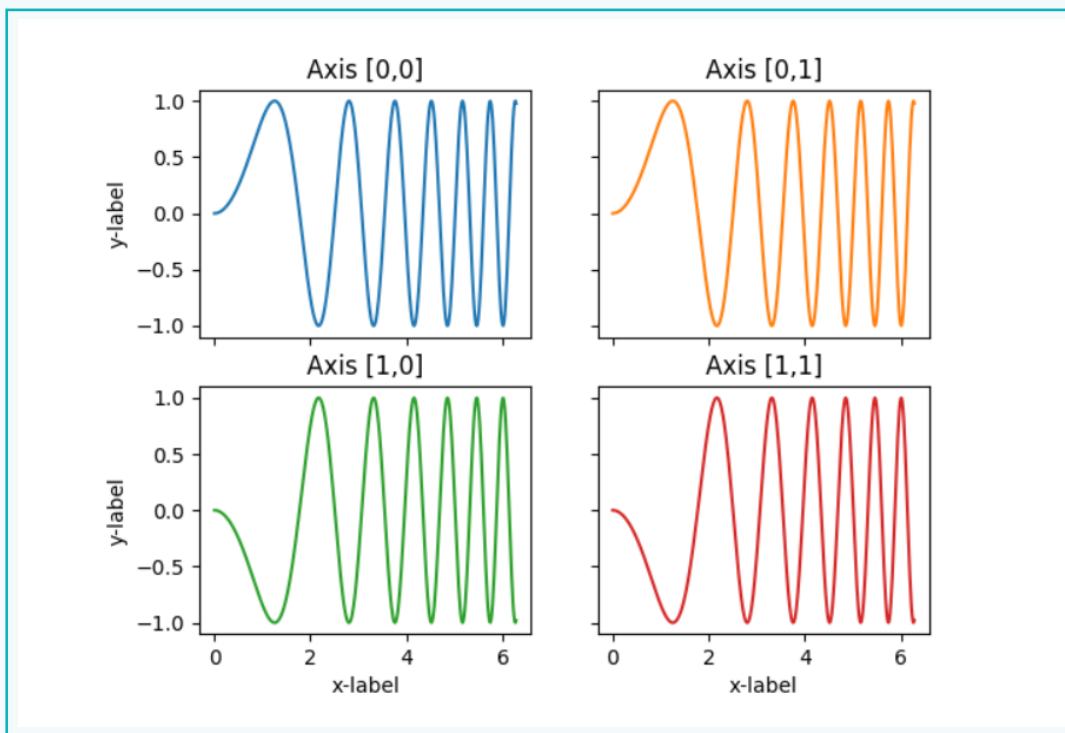
Activity

- 1 Use `plt.figure()` to create a new figure for plotting. Set the figure size to (12, 6) to provide enough space for the three plots.

```
19
20 # EDA
21 # Visualize relationships between marketing expenditures and sales
22 plt.figure(figsize=(12, 6))
23
...
```

Since we want to display and compare the relationships between all three marketing expenditures versus sales in a single figure, we will use subplots.

Subplot: A Subplot is a smaller plot within a larger figure or window. In Matplotlib, subplots are used to create multiple plots in a single figure, arranged in a grid-like layout. This allows for comparing different plots side by side or in a structured manner.



Subplot

Activity

Syntax:

`plt.subplot(nrows, ncols, index)`

Where,

- **nrows:** Number of rows in the subplot grid.
- **ncols:** Number of columns in the subplot grid.
- **index:** The index of the subplot to be created, starting from 1. It determines the position in the grid.

2 Create a Subplot for TV Expenditure vs Sales

- Use `plt.subplot()` to define the layout of subplots. Set (1,3,1) to create the first subplot in a 1-row, 3-column grid.
- Plot a scatter plot of TV expenditure versus sales using the `plt.scatter()` function. Use `df['TV']` to index the TV expenditure column and `df['sales']` to index the sales column from the DataFrame.
- Add a title with `plt.title()`, x-axis label with `plt.xlabel()`, and y-axis label with `plt.ylabel()`.

```
23
24 # TV vs Sales
25 plt.subplot(1, 3, 1)
26 plt.scatter(df['TV'], df['sales'])
27 plt.title('TV vs Sales')
28 plt.xlabel('TV Expenditure')
29 plt.ylabel('Sales')
30
```



3 Create a Subplot for Radio Expenditure vs Sales

- Use `plt.subplot()` to define the layout of subplots. Set (1,3,1) to create the second subplot in the same grid. This places the subplot in the second position.
- Plot a scatter plot of Radio expenditure versus sales using the `plt.scatter()` function. Use `df['radio']` to index the radio expenditure column and `df['sales']` to index the sales column from the DataFrame.
- Add a title with `plt.title()`, x-axis label with `plt.xlabel()`, and y-axis label with `plt.ylabel()`.

Activity

```
30
31 # Radio vs Sales
32 plt.subplot(1, 3, 2)
33 plt.scatter(df['radio'], df['sales'])
34 plt.title('Radio vs Sales')
35 plt.xlabel('Radio Expenditure')
36 plt.ylabel('Sales')
37
```

4 Create Subplot for Newspaper Expenditure vs Sales

- Use *plt.subplot()* to define the layout of subplots. Set (1,3,3) to create the third subplot in the same grid. This places the subplot in the third position.
- Plot a scatter plot of Newspaper expenditure versus sales using the *plt.scatter()* function. Use *df['newspaper']* to index the newspaper expenditure column and *df['sales']* to index the sales column from the DataFrame.
- Add a title with *plt.title()*, x-axis label with *plt.xlabel()*, and y-axis label with *plt.ylabel()*.

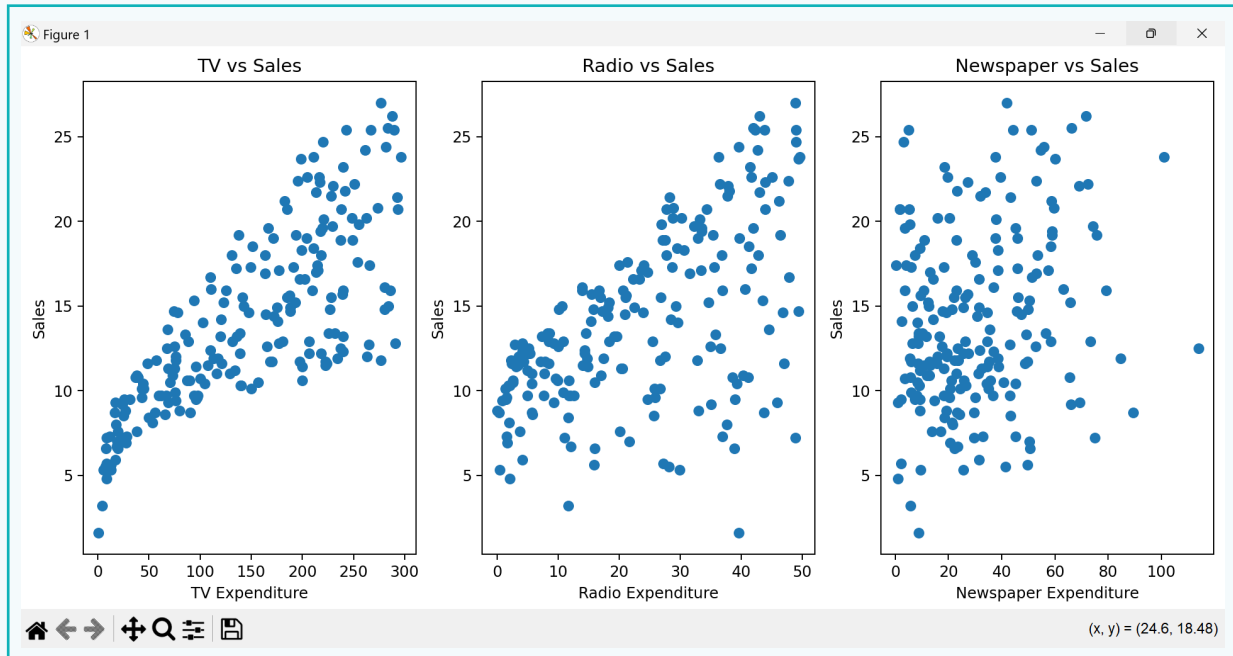
```
38 # Newspaper vs Sales
39 plt.subplot(1, 3, 3)
40 plt.scatter(df['newspaper'], df['sales'])
41 plt.title('Newspaper vs Sales')
42 plt.xlabel('Newspaper Expenditure')
43 plt.ylabel('Sales')
44
```

- Use *plt.tight_layout()* to adjust the spacing between subplots for a clean and non-overlapping layout.
- Display all the plots with *plt.show()*.

```
44
45 plt.tight_layout()
46 plt.show()
47
```

Activity

Output -



Now, analyse the plots to extract insights from them.

Findings:

- **TV expenditure** shows a strong positive link to increased sales, suggesting that investing more in TV advertising could significantly boost sales.
- **Radio expenditure** also correlates with higher sales, though its impact is less pronounced than TV ads. Radio can still be a valuable marketing tool.
- **Newspaper** expenditure has a weaker connection to sales, indicating it may not be as effective as TV or radio. Evaluating its cost-effectiveness or exploring other options might be necessary.

Next, we must provide a few suggestions based on our findings.

- **Increase TV ad spending** to maximise its strong impact on sales, while maintaining or slightly adjusting the current investment in radio advertising.
- **Reevaluate newspaper ad investments** as they demonstrate the weakest correlation with sales, suggesting a potential shift in budget allocation.

Activity

Part 5 - Build and Evaluate the Model

Create a Machine Learning model to predict future sales based on expenditure factors.

1 Define Features and Target

- Select the features (TV, radio, newspaper expenditures) and target variable (sales) for the model. Assign these to *X* and *y* variables, respectively.

```
56
57 # Build and Evaluate the Model
58 # Define features and target
59 X = df[['TV', 'radio', 'newspaper']]
60 y = df['sales']
61
```

2 Create and Train the Model

- Create a variable, name it as '*model*'. Initialise the *LinearRegression* model and train it using the *fit()* function with the features and target variable.

```
62 # Create and train the model
63 model = LinearRegression()
64 model.fit(X, y)
65
```

3 Make Predictions

- Create a new *DataFrame* with hypothetical expenditure values and save it in the '*new_expenditure*' variable.

```
65
66 # Make predictions
67 new_expenditure = pd.DataFrame({
68     'TV': [250],
69     'radio': [40],
70     'newspaper': [100]
71 })
72
```

Activity

- 4 Use the model to predict sales based on these values in *new_expenditure* and save it in *predicted_sales* variable. Print the *predicted_sales* to see the model's output.

```
72  
73 predicted_sales = model.predict(new_expenditure)  
74 print("Predicted Sales for new expenditures:", predicted_sales[0])  
75
```

Output -

```
Predicted Sales for new expenditures: 21.81750210578936  
PS D:\MY PYTHON PROJECTS>
```

Thus, we have successfully analysed the impact of marketing expenditures on sales, visualised key relationships, and built a machine-learning model to guide future advertising investments.

In this lesson, we explored the concept of Data Science, gaining an understanding of its project cycle and various applications. We performed an activity where data science principles were applied to solve a problem using real-world data.

Exercise

Fill in the Blanks.

- 1 Data Analysis is the method of studying or analysing data to find useful _____ and make informed _____.
- 2 Data Science combines data analysis with advanced methods of _____ to provide more accurate predictions and personalised _____.
- 3 The _____ Science Life Cycle includes phases such as defining the problem, collecting data, data cleaning, exploratory data analysis, and model _____.

State if the statements are True or False.

- 4 The data science life cycle includes phases such as defining the problem, collecting data, data cleaning, and model building.
 _____
- 5 In model building and deployment phase, all the necessary information to address the problem is gathered.
 _____
- 6 Exploratory Data Analysis (EDA) involves creating visualisations and applying basic statistical methods to understand the data better.
 _____

Tech Flash

- | | | |
|--|---|--|
| Data Analysis | : | The process of examining data to uncover patterns, trends, and insights for informed decision-making. |
| Data Science | : | The study of data using advanced methods and algorithms to extract meaningful insights and make predictions. |
| Exploratory Data Analysis (EDA) | : | An approach to analysing data sets to summarise their main characteristics and uncover patterns before formal modelling. |
| Scatter Plot | : | A graph that uses dots to represent the values of two different variables and reveals their relationship. |